Venkata Lakshmi Parimala Pasupuleti

Business Intelligence · AI Analytics · Python · SQL · AWS · Machine Learning · Automated Reporting · Executive Decision Support

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PROFESSIONAL SUMMARY

Senior Business Intelligence Engineer with 8+ years building **AI analytics tools** and **automated reporting** pipelines that drive **executive decision support** at scale. Built LeaseI — a production AI platform delivering **actionable insights** to non-technical stakeholders from complex compliance data: document intelligence, root-cause investigation, and self-service analytics with citation-grounded outputs. At Entrata (PropTech SaaS, 100K+ MAU), owned **data extraction, transformation, and cleaning** across 1.6M+ records using advanced SQL and Python automation pipelines — 200+ queries tuned for 30% performance gain, 80% manual workload eliminated. Built AgentProbe — an open-source **AI-based tool** for automated evaluation of machine learning and agentic systems. Strong in **AWS (S3, EC2, RDS)**, distributed data processing (Apache Spark/PySpark, Hadoop-compatible paradigm), **machine learning**, data mining, and cloud storage and computing. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Business Intelligence (BI) | AI Analytics & AI-Based Tools | Automated Reporting & Operational Metrics | Executive Decision Support | Python Scripting & Automation | Advanced SQL & Data Extraction | AWS (S3, EC2, RDS, Redshift) | Hadoop / Apache Spark / PySpark | Machine Learning & Data Mining | Information Retrieval & RAG Pipelines | Data Integrity & Validation | Self-Service Analytics | Scalable Data Pipelines & ETL | Cross-Functional Stakeholder Collaboration

EXPERIENCE

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Research Associate · AI Analytics · Data Science · Executive BI Dashboards

- Built multi-source **data science and machine learning** pipelines (Python, R) on WHO and UK Road Safety datasets — **data extraction, transformation, and cleaning** across 190+ countries achieving $R^2 = 0.977$; statistical analysis and hypothesis testing driving **actionable insights** for faculty decision-making.
- Designed **executive-level BI dashboards** (Tableau, ggplot2, Quarto) delivering **automated reporting** of operational metrics across two research cohorts — improved stakeholder comprehension by 40%; applied Isolation Forest **anomaly detection** to automate **data integrity** checks across 3 pipelines.
- Delivered technical instruction to 600+ students in Python, SQL, database design, and **data mining** techniques; provided targeted analytical support and rubric-aligned feedback.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Data Engineering · SQL Optimization · AWS · Automated Pipelines

- **Extracted, transformed, and cleaned large datasets** at scale: designed SQL import scripts processing 1.6M+ HUD and Tax Credit income records across 5 US states; built **ETL pipelines** for financial data (HAP allocation, rent pricing, HUD payments, tax credit calculations) using Python, PostgreSQL, and **AWS S3** — enforcing **data integrity** and validation across all client environments.
- **Automated reporting and operational metrics**: eliminated 80% of manual operational workload via Python and Bash **scripting for automation**; built occupancy risk dashboards and recertification compliance metrics tracking 100K+ unit portfolios; **self-service analytics** tools enabling compliance and operations teams to access data independently without engineering support.
- **Optimized 200+ SQL queries** across all client environments (30% performance gain) through execution plan tuning, indexing, database partitioning, caching, and audit tables; led **scalable** database architecture decisions for complex production schemas in collaboration with data architects; worked with **AWS (S3, EC2, RDS)** in production — 100% PHI compliance migration, zero security incidents.
- Led **microservices architecture** redesign using Docker — 50% scalability improvement, 30% deployment time reduction; built **cloud computing** infrastructure converting 20% of product modules to service-based architecture; mentored 5+ engineers in data engineering best practices; conducted 300+ peer code reviews.
- Built **Salesforce integrations and CRM data pipelines** (Clay, n8n) connecting product data to commercial systems; automated **cross-functional** reporting workflows for business stakeholders across compliance, operations, and product teams.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — AI Analytics Pipeline · Automated Decision Support

- Built an agentic **AI analytics pipeline** (LangGraph stateful graph) that automatically derives **actionable insights** from operational data: applies **machine learning**-driven decision logic, extracts structured outputs, and routes results with cited, grounded recommendations — designed for **non-technical stakeholders** to consume without engineering support.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — ETL · SQL · AWS S3 · Data Pipelines

- Built affordable housing **data pipelines** (PHP, PostgreSQL); designed SQL **data extraction** scripts processing 1.6M income records; migrated thousands of client documents to **AWS S3**; redesigned document storage database architecture for **scalable** retrieval; created 56 state-specific compliance documents with automated merge fields.

PORTFOLIO PROJECTS

Leasel — AI-Based Analytics Platform for Non-Technical Business Stakeholders

AI Tools · Automated Reporting · Actionable Insights

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, Information Retrieval, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a 3-surface **AI-based tool** for compliance operations teams (non-technical stakeholders): (1) Document Intelligence — automated **information retrieval** from 1,240 lease documents with 100% citation correctness; (2) Support Investigation — 8-agent **AI analytics** pipeline deriving ranked, evidence-backed root-cause insights from business rules and historical incidents; (3) Self-Service — **self-service analytics** layer for tenant policy queries with safety boundary enforced in code.
- Achieved **93.6% precision** and 87.9% F1 across 10 compliance rules (5 US states + federal, 1,050 synthetic records); **data integrity** enforced at every layer: LLM extracts and reasons, deterministic logic validates — 0% unsupported-claim rate; designed to **scale** across multiple business units sharing one knowledge repository.

AgentProbe — AI-Based Automated Evaluation Engine for Machine Learning Systems

AI Analytics · Automation · Scalable Processes

2024–Present · Python, LangGraph, FastAPI, ChromaDB, OpenAI, Anthropic, Groq, Ollama, React, Docker, GitHub Actions · github.com/venkatalakshmiparimala/AgentProbe

- Built an open-source **AI-based tool** for **automated** evaluation of machine learning and agentic systems: 5-metric eval engine (factual accuracy, hallucination, retrieval, completeness, compliance), **scalable** multi-provider LLM routing, CI-integrated regression gating via GitHub Actions — 100% local, zero cost.
- 15-endpoint FastAPI backend with 3-agent LangGraph runtime and trace-level observability dashboard tracking per-run **operational metrics** and drift — designed for **non-technical stakeholders** to monitor AI system performance without engineering involvement.

CrashClarity — Large-Scale Data Mining & Predictive Analytics (104K Records)

Data Mining · Machine Learning · BI Dashboard

2025 · Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering, Tableau, GitHub Pages · View Project

- End-to-end **data mining** and **machine learning** pipeline on 104K+ UK accident records — **data extraction, transformation, cleaning**, feature engineering across behavioral and environmental variables; 85%+ model accuracy with Random Forest + Logistic Regression; interactive **BI dashboard** deployed on GitHub Pages delivering **actionable insights** to cross-functional stakeholders.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

Coursework: Machine Learning, Data Mining, Statistical Analysis, Information Retrieval, Data Science, Cloud Computing, Applied Analytics

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

BI & Analytics: Business Intelligence, AI Analytics, Automated Reporting, Operational Metrics, Executive Decision Support, Self-Service Analytics, Actionable Insights, Data Science, Data Mining, Information Retrieval, Anomaly Detection

Data & Query: Advanced SQL (PostgreSQL, MySQL, Oracle), Python (Pandas, NumPy, scikit-learn, PySpark), R, ETL, Data Extraction, Data Transformation, Data Cleaning, Data Integrity, Scalable Data Pipelines, Apache Spark, Hadoop (Spark-compatible), AWS Redshift (PostgreSQL-compatible)

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Cloud Storage, Cloud Computing, Docker, CI/CD (GitHub Actions), Microservices, Distributed Systems

ML & AI: Machine Learning, Random Forest, Logistic Regression, Isolation Forest, XGBoost, Feature Engineering, Model Evaluation, LangChain, LangGraph, RAG Pipelines, ChromaDB, Embeddings, NLP, BERT, spaCy, Hugging Face, PyTorch, scikit-learn

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, ggplot2, Quarto, Dashboard Design

Languages: Python (primary), SQL, R, JavaScript, TypeScript, PHP, PL/SQL, Bash